B. TECH-ME-III-II Sem

23A03604e	SMART MATERIALS (Professional Elective-II)	L	T	P	C
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Course objectives: The objectives of the course are to	
1	Fundamental characteristics of different metals and provide an understanding of smart materials, their classification and real-world applications.
2	Knowledge of various types of smart materials and electro rheological fluids, and shape memory materials.
3	Processing techniques of various smart materials, and smart fluids, with a focus on synthesis and fabrication methods such as metallization and UV curing.
4	Understanding of various types of sensors, and advanced sensors such as carbon nanotube and polymer-based sensors.
5	Principles, types, and applications of actuators used in smart systems, and electro thermal actuators.

COURSE OUTCOMES On successful completion of this course the student will be able to		
1	Understand and distinguish between traditional engineering materials and smart materials, and identify appropriate smart materials for various engineering applications.	*L1,L2,L3*
2	Explain the working principles, properties, and applications of different smart materials and evaluate their suitability for specific engineering and technological applications.	L2,L3,L5
3	Understand and apply suitable processing and fabrication techniques for different smart materials in engineering applications.	L1,L2,L3
4	Identify, describe, and equate different sensor technologies and select appropriate sensors for engineering applications.	L1,L2,L5
5	Demonstrate the working mechanisms of various actuators compare and select suitable actuation methods for different smart materials for create system applications.	L2,L3,L6

UNIT-1

Introduction

Characteristics of metals, polymers and ceramics. Introduction to smart materials. Classification of smart materials, Components of a smart System, Applications of smart material.

UNIT-2

Smart Materials Piezoelectric materials, Electro strictive Materials, Magnetostrictive materials, Magnetoelectric materials, Magnetorheological Electrorheological fluids, Shape Memory materials.

UNIT-3

Processing of Smart Materials Semiconductors and their processing, Metals and metallization techniques, Ceramics and their processing, Polymers and their synthesis, UV radiation curing of polymers, fluids.

UNIT-4

Sensors Introduction, Conductometric sensors, Capacitive sensors, Piezoelectric sensors, Magnetostrictive sensors, Piezoresistive sensors, Optical sensors, Resonant sensors, semiconductor-based sensors, Acoustic sensors, polymerize sensors, Carbon nanotube sensors.

UNIT-5

Actuators Introduction, Electrostatic transducers, Electromagnetic transducers, Electrodynamic transducers, Piezoelectric transducers, Electro-strictive transducers, Magneto-strictive transducers, Electro thermal actuators, Comparison of actuation, Applications

Text Books:

1. Smart Material Systems and MEMS: Design and Development Methodologies, V. K. Varadan, K. J. Vinoy, S. Gopalakrishnan, John Wiley and Sons, England, 2006.

2. Smart Structures and Materials, Brain Culshaw, Artech House, London, 1996.

3. Smart Materials and Structures, Mukesh V. Gandhi, Brian S. Thompson, , Springer, May-1992.

Reference Books:

1. Smart Structures: Analysis and Design, A. V. Srinivasan, Cambridge University Press, Cambridge, New York, 2001.
2. Smart Structures, P. Gauenzi, Wiley, 2009.
3. Piezoelectric Sensorics: Force, Strain, Pressure, Acceleration and Acoustic Emission Sensors, Materials and Amplifiers, G. Gautschi, Springer, Berlin, New York, 2002.
4. Analysis and Performance of Fiber Composites, B. D. Agarwal and L. J. Broutman, John Wiley & Sons.
5. Engineering aspects of Shape memory Alloys, T. W. Duerig, K. N. Melton, D. Stockel,C.
6. Mayman, Butterworth – Heinemann, 1990.

Web Resources:

1. https://nptel.ac.in/courses/112104173/

2. www.iop.org/EJ/article/0964-1726/5/3/002/sm6301.ps.gz

MOOCs:

1. https://nptel.ac.in/courses/112104173/

2. https://nptel.ac.in/courses/112104251/